TASK 9 – JUNIT TESTING

PROBLEM STATEMENT:

Develop a **Java Calculator application** and perform **unit testing using JUnit5**.

The Calculator class should provide the following operations:

1. **Addition** – Add two integer numbers.
2. **Division** – Divide one integer by another.
3. **Division by Zero Handling** – Throw an IllegalArgumentException when division by zero is attempted.

Create a **JUnit 5 test class named CalculatorTest** to verify these operation :

assertEquals() to test addition and division.

assertThrows() to test division by zero.

Run the test cases in Eclipse using **Run As → JUnit Test** and verify that all test cases pass successfully. Also demonstrate a **failed test case** by providing an incorrect expected output and observe the failure in the JUnit result window.

SOURCE CODE:

Calculator code: **package**

junitDemo;

```
public class Calculator {

    public int add(int a, int b) {

        return a + b;

  }
```

```
   public int divide(int a, int b) {        if (b == 0) {            throw
new IllegalArgumentException("Division by zero");
       }

       return a / b;
   }
}
```

CalculatorTest code:

```
package junitDemo;

import static org.junit.jupiter.api.Assertions.*;

import org.junit.jupiter.api.Test;

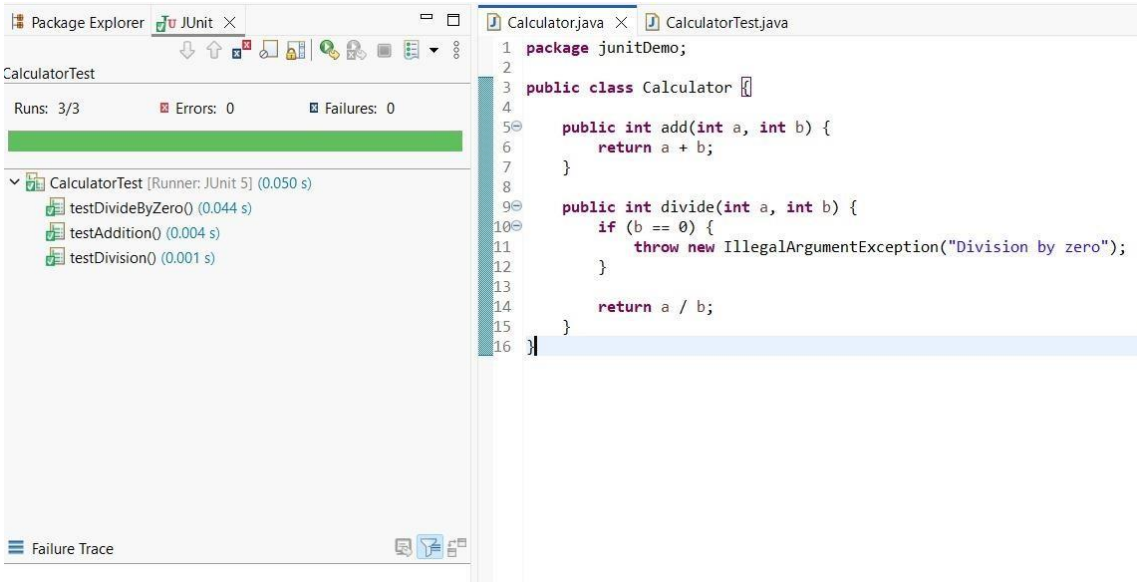
public class CalculatorTest {

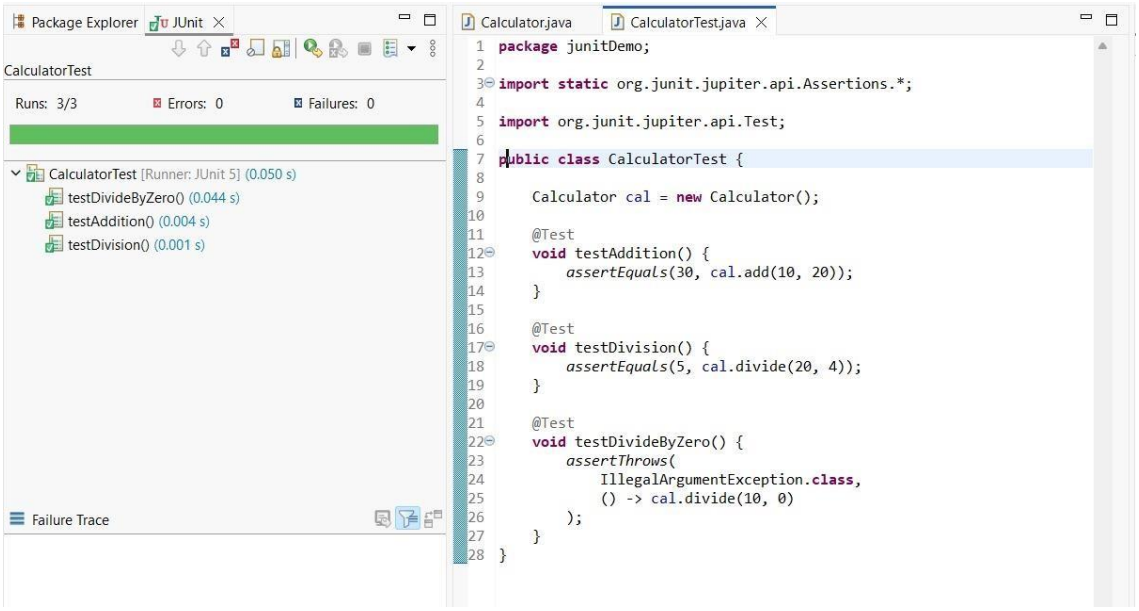
   Calculator cal = new Calculator();

   @Test     void testAddition() {
assertEquals(30, cal.add(10, 20));     }
```

```
    @Test     void testDivision() {
assertEquals(5, cal.divide(20, 4));
    }
    @Test     void
testDivideByZero() {
assertThrows(
            IllegalArgumentException.class,
            () -> cal.divide(10, 0)
        );
    }
}
```

OUTPUT:





Code:

```
package junitDemo;

import static org.junit.jupiter.api.Assertions.*;

import org.junit.jupiter.api.Test;

public class CalculatorTest {

   Calculator cal = new Calculator();

   @Test
   void testAddition() {
      assertEquals(30, cal.add(10, 20));
   }

   @Test
   void testDivision() {
```

```
        assertEquals(5, cal.divide(20, 4));
    }

    @Test
    void testDivideByZero() {
        assertThrows(
            IllegalArgumentException.class,
            () -> cal.divide(10, 0)
        );
    }
}
```